

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12.	(a)	(i)		The iodide ions (moved to the anode and) were oxidised / lost electrons forming iodine (1) $2\text{I}^-(\text{aq}) \rightarrow \text{I}_2(\text{aq}) + 2\text{e}^- \quad (1)$ (Aqueous) iodine was produced giving a yellow/brown coloration (around the anode) (1)	1	1	1	3		1
		(ii)		Both iodide and chloride ions formed precipitate – yellow and white observed (1) On adding ammonia and shaking the white precipitate / silver chloride dissolves (1) Leaving silver iodide as a pale yellow solid / silver iodide does not dissolve (1)		3		3		3
		(iii)		Number of moles of potassium iodate(V) = $\frac{0.100 \times 18.00}{1000}$ = 1.8×10^{-3} (1) Number of moles of NaI present in 25.0 cm³ of the solution of the mixture = $1.8 \times 10^{-3} \times 2 = 3.6 \times 10^{-3}$ ∴ Number of moles of sodium iodide in 250 cm³ = 0.036 (1) Mass of sodium iodide present in 250 cm³ of the solution of the mixture = $0.036 \times 150 = 5.40$ g ∴ % sodium iodide in the mixture = $\frac{5.40 \times 100}{11.24} = 48$ (1) ecf possible		3		3	1	

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12.	(b)			$f = c/\lambda = 3 \times 10^8 / 278 \times 10^{-9} = 1.079 \times 10^{15} \text{ (Hz)}$ (1) $E = hf \quad \therefore E = 6.63 \times 10^{-34} \times 1.079 \times 10^{15} \text{ J}$ $= 7.154 \times 10^{-19} \text{ J (per molecule)}$ $\therefore \text{per mole} \quad 7.154 \times 10^{-19} \times 6.02 \times 10^{23} \text{ J mol}^{-1}$ $= 4.307 \times 10^5 \text{ J mol}^{-1} \text{ (1)}$ $= 431 \text{ kJ mol}^{-1}$			2	2	2	
	(c)			As the group is descended the bond energies decrease and the wavelengths increase / astatine is below iodine in the Periodic Table / $\lambda_{\text{max}} > 400\text{nm}$ (1) coloured gas linked with the visible region (1)			2	2		
				Question 12 total	1	7	5	13	3	4